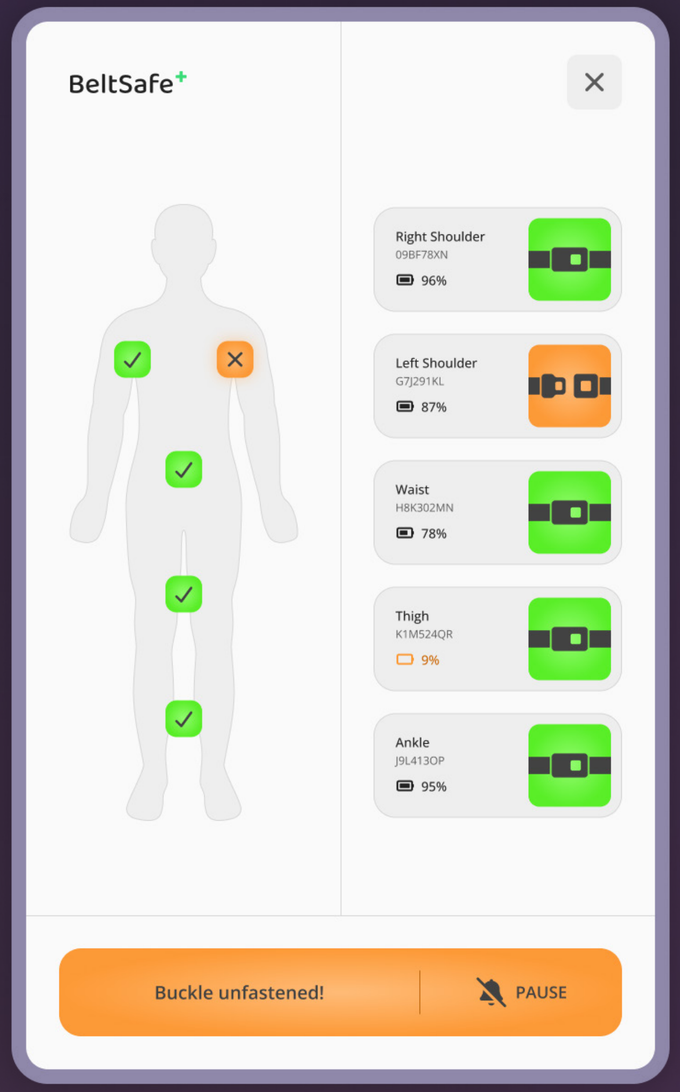
I love making things.

Interaction, interface & experience things.

Digital things, physical things; early-stage things and production-ready things.

Things that do good.

Things that put a smile on your face.



Protecting patients in the back of ambulances

ORTUS TECHNOLOGIES — IN PROGRESS

THE CHALLENGE

Beltsafe was all about creating a retrofit system that notifies ambulance crews if their on-board stretcher is unbuckled and therefore their patient at risk.

A system that could cope with stretchers changing throughout the day. Deal with the abuse of a high-stake environment where lives are at risk. Last a year in the field. Avoid overwhelming very stressed crews.

Design of interactions in a hybrid digital-physical system including Android app for the crew and LED-based hardware display for the drivers. Collaboration with developers and contribution to live code, closing the gap between design intent and shipped product.

KEY ACTIVITIES

// UX/UI design of hardware display and app

// Definition of interaction logic across all parts of the system

// Front-end code contribution to production app (React Native)

// 3D CAD, 3D printing and assembly for driver hardware display

BeltSafe+ Insights

Summary

Vehicles with **disengaged** buckles **5 / 10**

Vehicles with **active alarms** **2 / 10**

Vehicles with **low battery** buck

Fleet List

VEHICLE REG.	BUCKLE STATUS	ALARM STATUS	BATTERY STATUS	LAST UPDATED
KL25 THG	3 / 5	Active	–	2 mins ago
KY25 ZAB	3 / 5	Muted	–	3 mins ago
BR25 STV	5 / 5	–	–	1 day ago
MN25 OQS	5 / 5	Muted	–	2 mins ago
GH75 IJK	3 / 4	Active	–	2 mins ago
VH25 XPZ	5 / 5	–	–	2 mins ago
CD25 EFG	5 / 5	–	2 Low	2 mins ago
HJ75 KUL	3 / 3	–	–	2 mins ago
KU75 CPE	5 / 5	Muted	–	2 mins ago
LM25 NMP	5 / 5	–	–	2 mins ago





Transforming legacy scientifc control software

THERMAL HAZARD TECH — 2025

THE CHALLENGE

A feeling of focus, precision and a sense of command. The same feeling you imagine as a kid at the controls of a spaceship, darting from galaxy to galaxy. Whimsical, yes, but an incredibly powerful experience design reference too.

Having redesigned a lot of legacy software, with THT and their calorimeters, we found ourselves thinking less about the person at the controls and more about what it feels like to be in command of something powerful.

A ground-up rebuild of a design system and core desktop user interfaces. A fully-fledged colour, typography and layout system encapsulated in a real-code prototype. Made to stand apart from the competition, and to prove serious tools can still feel like play.

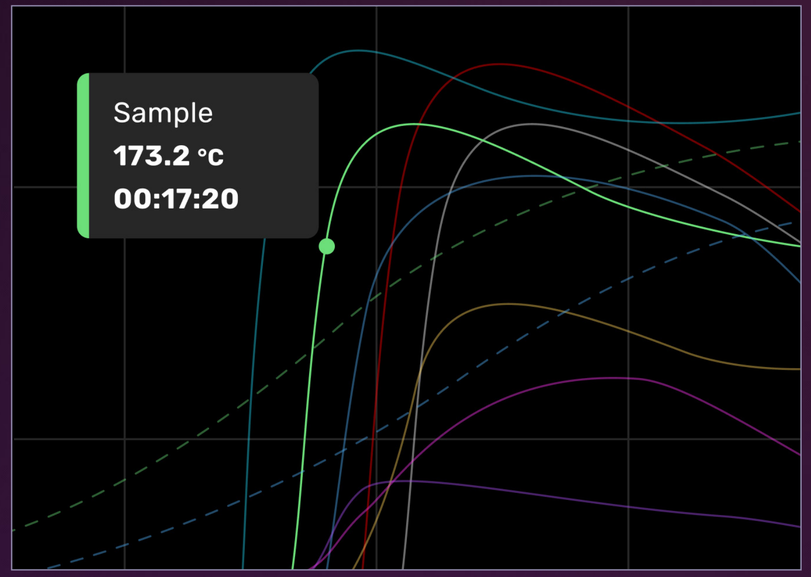
KEY ACTIVITIES

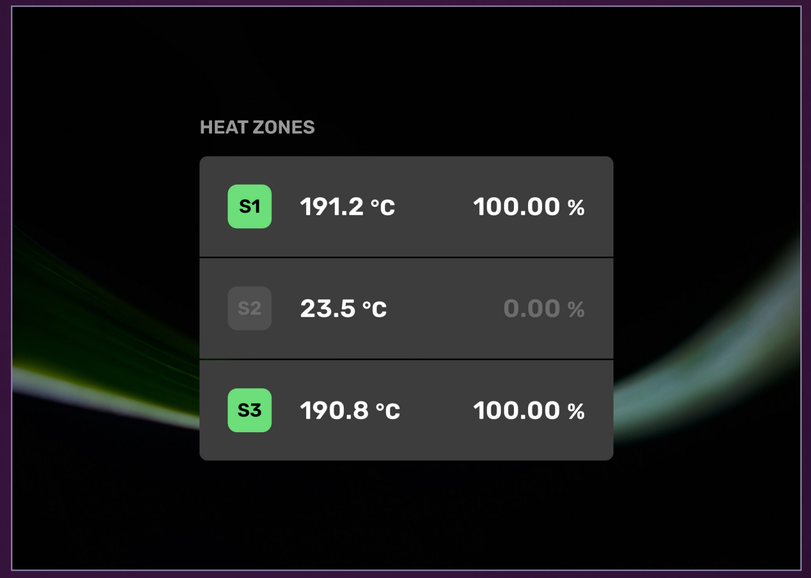
// User research and testing with a community of scientists

// Owning creation of UX, UI and design system

// Prototyping-in-code (React)


ARCES / HEAT-WAIT-SEEK
TestName
TEMPERATURE
21.0 °C
TEMPERATURE
+ 0.50 °C/min
PRESSURE
1.20 bar
PRESSURE
– 0.02 bar/min
Setup
Start Test
CURRENT MODE
Isothermal
System
Core
Auxiliary
T 190.0 °C 100.00 %
S1 190.0 °C 100.00 %
S2 21.0 °C 0.00 %
S3 190.0 °C 100.00 %
S4 21.0 °C 0.00 %
S5 21.0 °C 0.00 %
S6 190.0 °C 100.00 %
B 21.0 °C 0.00 %
Data
System Data
Test Data
Messages
GRAPH KEY
Temperature (°C)
Pressure (bar)
Elapsed Time (HH:MM)
0
50
100
150
200
250
1.00
1.20
1.40
1.60
1.80
2.00
00:10
00:20
00:30
00:40


Sample
173.2 °C
00:17:20


HEAT ZONES
S1 191.2 °C 100.00 %
S2 23.5 °C 0.00 %
S3 190.8 °C 100.00 %

Empowering clinicians to create next-gen medicines

MFX 2025 – 2026

THE CHALLENGE

MFX are pioneering novel cell and gene therapy manufacturing, building the hardware and software scientists need to scale life-saving treatments from lab experiment to clinical production.

Nearly 100 requirements to work through. A complex, novel technical system where things can and do go wrong for early adopters. Users who need to feel confident in the state of their system, and know exactly how to fix it when something breaks.

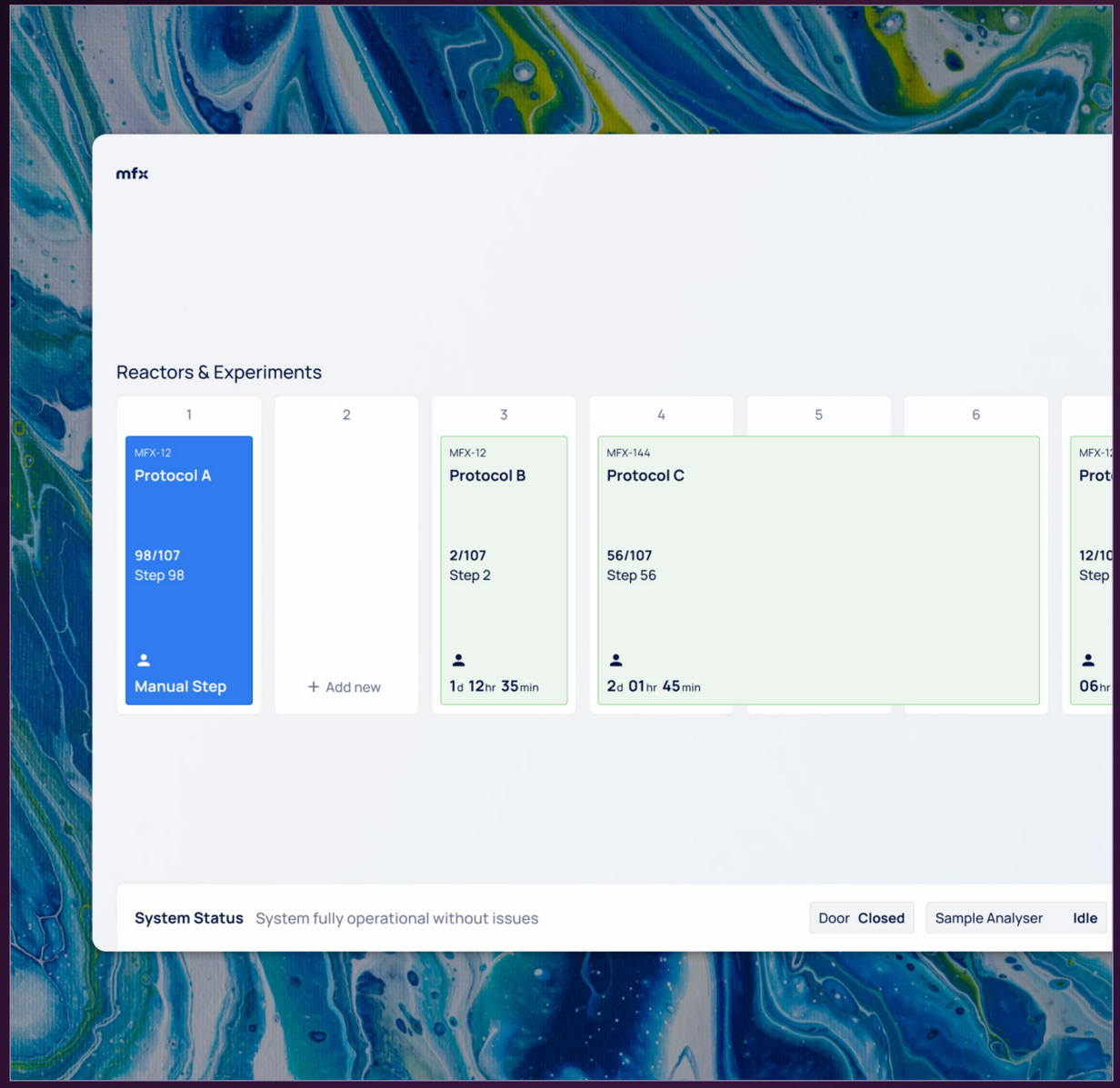
Prioritisation against MFX's hardware roadmap to focus on what mattered first. Interaction flows mapped to catch every edge case a user might hit. A new UI kit built on MUI, giving scientists interactive controls alongside real-time data visualisation. Iterating towards great outcomes in collaboration with developers and biologists.

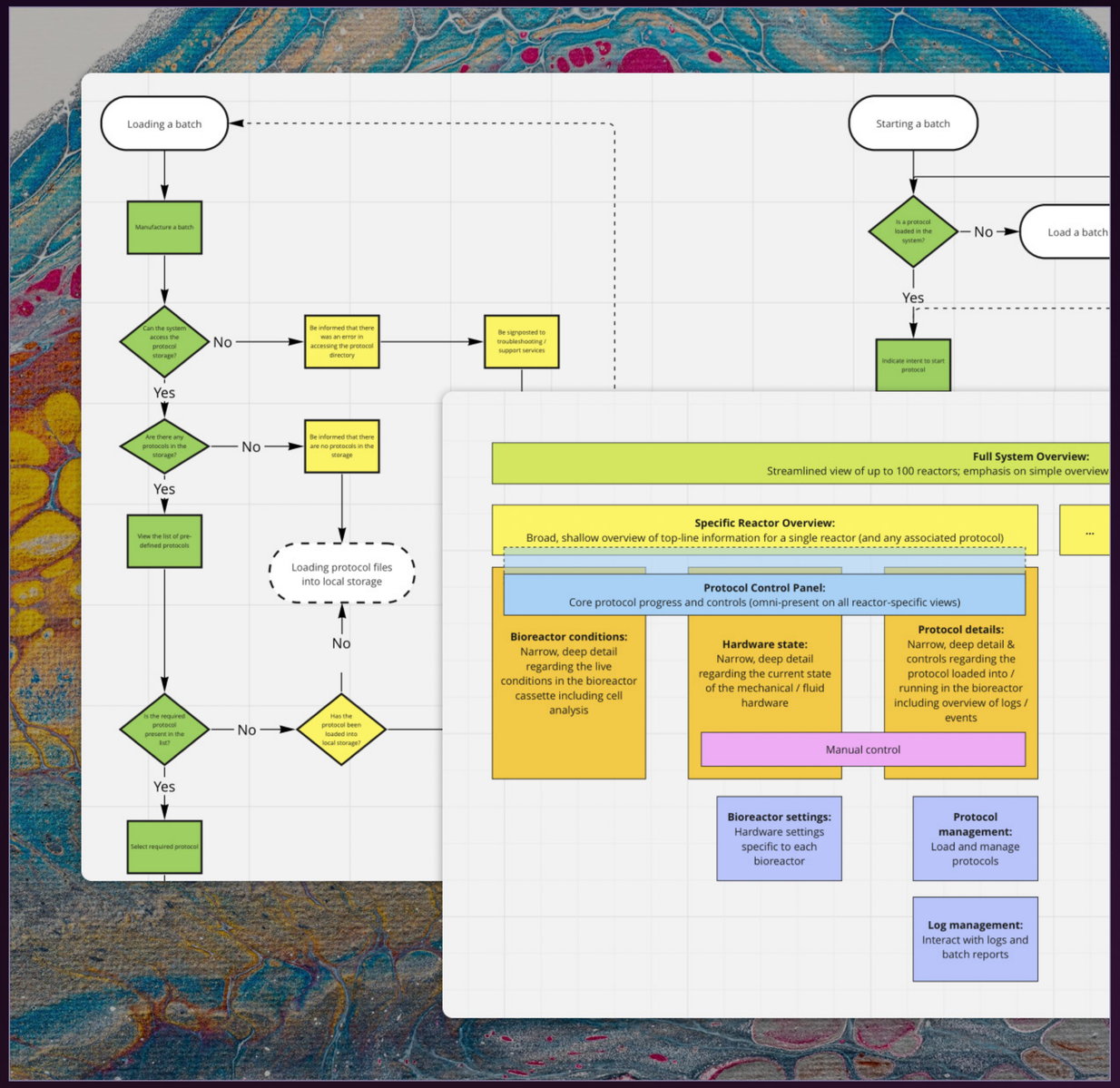
KEY ACTIVITIES

// UX + UI design of hardware control and monitoring interfaces

// Research, triage and prioritisation of product requirements

// Creation of design system React-based UI kit, built on MUI base




Loading a batch
Manufacture a batch
Can the system access the protocol storage?
No
Be informed that there was an error in accessing the protocol directory
Be signposted to troubleshooting / support services
Yes
Are there any protocols in the storage?
No
Be informed that there are no protocols in the storage
Yes
View the list of pre-defined protocols
Loading protocol files into local storage
No
Is the required protocol present in the list?
No
Has the protocol been loaded into local storage?
Yes
Select required protocol
Starting a batch
Is a protocol loaded in the system?
No
Load a batch
Yes
Indicate intent to start protocol
Full System Overview:
Streamlined view of up to 100 reactors; emphasis on simple overview
Specific Reactor Overview:
Broad, shallow overview of top-line information for a single reactor (and any associated protocol)
...
Protocol Control Panel:
Core protocol progress and controls (omni-present on all reactor-specific views)
Bioreactor conditions:
Narrow, deep detail regarding the live conditions in the bioreactor cassette including cell analysis
Hardware state:
Narrow, deep detail regarding the current state of the mechanical / fluid hardware
Protocol details:
Narrow, deep detail & controls regarding the protocol loaded into / running in the bioreactor including overview of logs / events
Manual control
Bioreactor settings:
Hardware settings specific to each bioreactor
Protocol management:
Load and manage protocols
Log management:
Interact with logs and batch reports


Dissolved Oxygen
Full Protocol
Seeding
Activation
Transduction
Expansion
Dissolved Oxygen [%]
110.0
105.0
100.0
95.0
90.0
85.0
80.0
75.0
70.0
65.0
0:00
2:00
4:00
6:00
8:00
10:00
12:00
Time [hrs]
95.3 %
Transduction Phase
16-Jan-2023 at 10:07
Upcoming Interactions
Cassette 2 CGT-EXP-1 User interaction required
Cassette 8 CGT-EXP-1 11 mins
Cassette 9 CGT-EXP-1 52 mins
Cassette 3 CGT-EXP-3 3 hr 15 mins
Cassette 12 CGT-EXP-3 4 hr 23 mins
Cassette 21 CGT-EXP-2 1 day 1 hr 36 mins
Microscope Imaging
Confluence Graph
Image taken at 10:02
Confluence 19.2 %
0:00
3:00
6:00
9:00
12:00

Let’s build!

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